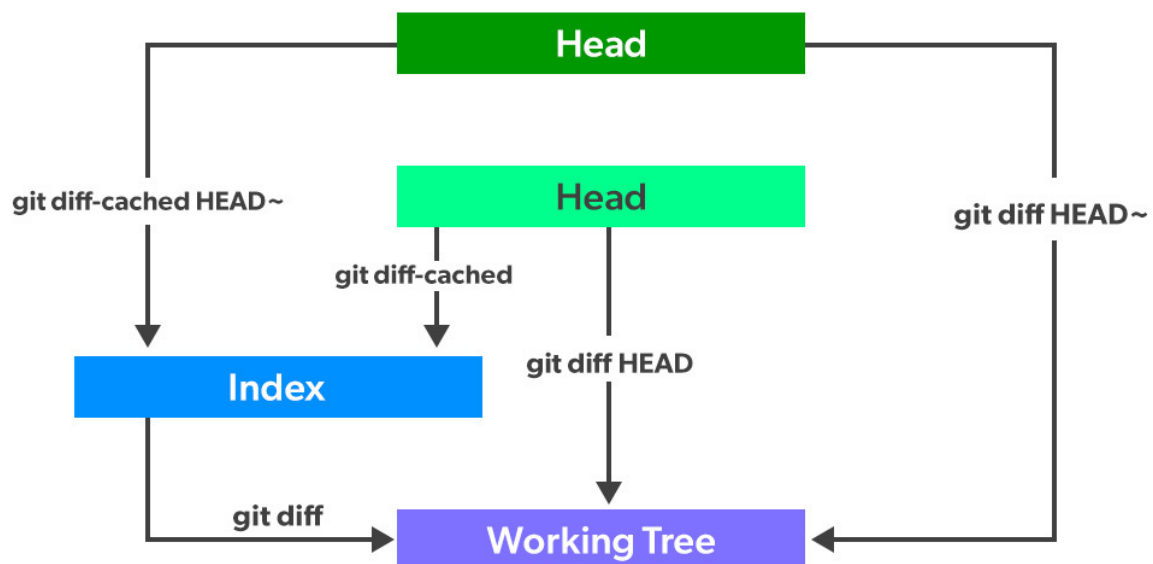


Industrial Tool

Git & Github

Git Diff :-

- git diff shows changes made in a Git repository.
- It compares different states of files.
- It shows differences between the working directory and staging area.
- It shows differences between the staging area and last commit.
- It can compare any two commits.
- It helps developers review changes before committing.



- The diagram shows the three main Git areas: HEAD, Index (staging area), and Working Tree.
- HEAD represents the latest commit.
- Working Tree contains your current files.
- Index holds staged changes.
- git diff shows unstaged changes (Working Tree ↔ Index).
- git diff HEAD shows all local changes (Working Tree ↔ HEAD).
- git diff --cached shows staged but uncommitted changes (Index ↔ HEAD).
- Arrows indicate how each git diff command compares these stages.

Viewing Changes Between Commits :-

- git diff shows changes made after the last commit.
- It displays differences between the committed version and the modified file.
- For example, if a.txt is changed after a commit, git diff shows those changes.
- It helps you see what exactly was modified in the file.
- These changes are shown before staging or committing again.

```
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ vi at.txt

Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   at.txt

no changes added to commit (use "git add" and/or "git commit -a")

Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff
warning: LF will be replaced by CRLF in at.txt.
The file will have its original line endings in your working directory
diff --git a/at.txt b/at.txt
index c9e425e..db75356 100644
--- a/at.txt
+++ b/at.txt
@@ -1,1 @@
-Hello Geeks for Geeks 😊
+Hello Geeks for Geeks 😊
```

Using git diff

Illustration: Changes between two Commits as shown below as follows:

- To see changes between two commits, use: git diff commit-id1 commit-id2
- This command compares one commit with another.
- It shows what changed from the first commit to the second commit.
- In the first commit, the text was "Hello Geeks for Geeks ?".
- In the second commit, the updated content is shown.
- This helps track how a file evolved between commits.

```
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff d4d7e49 eade69d
diff --git a/at.txt b/at.txt
index c9e425e..db75356 100644
--- a/at.txt
+++ b/at.txt
@@ -1,1 @@
-Hello Geeks for Geeks 😊
+Hello Geeks for Geeks 😊
```

Here can I use the command git diff commit-id-1 commitid-2

Shows difference for Staged files :-

- To see changes between the previous commit and staged files, use: `git diff --staged`
- It shows staged but uncommitted changes.
- `git diff --cached` does the same thing as `--staged`.
- Both compare the Index (staging area) with HEAD.
- `git status -v` also shows staged changes in detail.
- It works as a similar alternative for viewing staged differences..

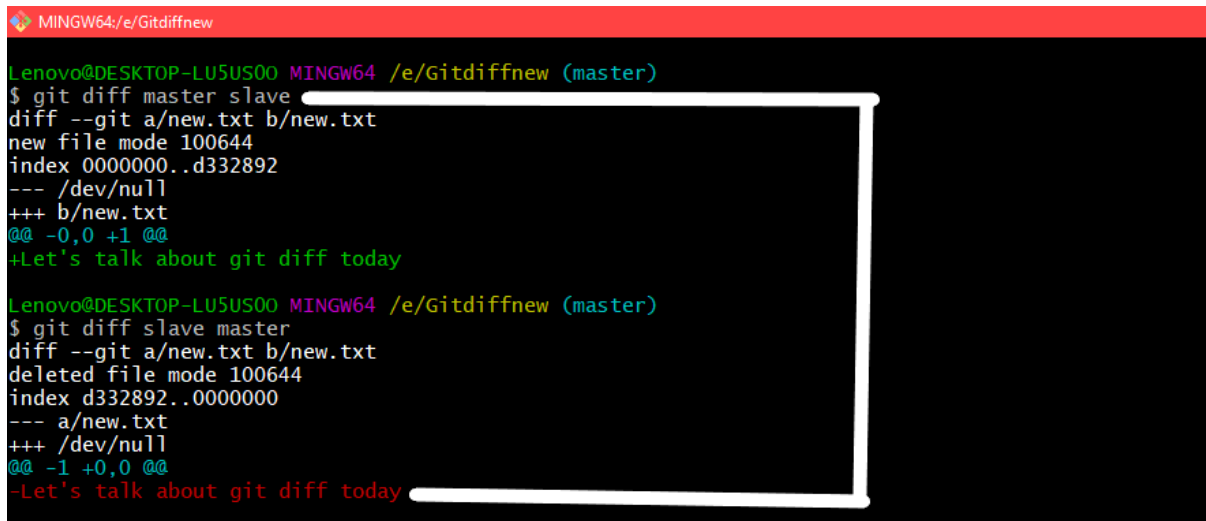
A terminal window with a red title bar that reads "MINGW64:/e/Gitdiffnew". The terminal shows the command `git diff --staged` being executed. The output indicates a new file `b/bw.txt` has been staged, comparing its index hash `0000000` to the previous commit hash `..88c801c`. The diff shows the file was previously empty (`--- /dev/null`) and now contains the text `hello git` (`+++ b/bw.txt`). A large white arrow points from the text "new file" towards the diff content. The prompt shows the user is on the `master` branch.

```
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff --staged
diff --git a/bw.txt b/bw.txt
new file mode 100644
index 0000000..88c801c
--- /dev/null
+++ b/bw.txt
@@ -0,0 +1 @@
+hello git
```

Using `git diff --staged`

Using git diff for comparing branches :-

- To compare changes between two branches, use:
git diff branch1_name branch2_name
- This command shows differences between branch1 and branch2.
- It displays all changes present in branch2 compared to branch1.
- It helps understand what will change when branches are merged.

A screenshot of a terminal window with a black background and a red title bar that reads 'MINGW64; e:/Gitdiffnew'. The terminal shows two git diff commands and their outputs. The first command is 'git diff master slave', which shows a new file 'b/new.txt' in green. The second command is 'git diff slave master', which shows the deletion of 'b/new.txt' in red. A white L-shaped bracket highlights the output of both commands.

```
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff master slave
diff --git a/new.txt b/new.txt
new file mode 100644
index 0000000..d332892
--- /dev/null
+++ b/new.txt
@@ -0,0 +1 @@
+Let's talk about git diff today

Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff slave master
diff --git a/new.txt b/new.txt
deleted file mode 100644
index d332892..0000000
--- a/new.txt
+++ /dev/null
@@ -1 +0,0 @@
-Let's talk about git diff today
```

Comparing the two branches by using the above command

- git diff branch1 branch2 → shows changes in branch2 not in branch1 (green).
- git diff branch2 branch1 → shows changes in branch1 not in branch2 (red).
- Green = changes present in the second branch.
- Red = changes missing from the base branch.
- Useful for quick branch comparison before merging.

Showing Both staged and unstaged Changes :-

- To see all staged and unstaged changes, use: `git diff HEAD`
- It compares the working directory and staging area with the last commit (HEAD).
- It shows everything changed after the last commit.

```
MINGW64/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff HEAD
warning: LF will be replaced by CRLF in d.txt.
The file will have its original line endings in your working directory
warning: LF will be replaced by CRLF in bw.txt.
The file will have its original line endings in your working directory
diff --git a/bw.txt b/bw.txt
new file mode 100644
index 0000000..36878e1
--- /dev/null
+++ b/bw.txt
@@ -0,0 +1 @@
+hello git users hello
diff --git a/d.txt b/d.txt
index 93541f9..8bcd78 100644
--- a/d.txt
+++ b/d.txt
@@ -1,2 +1,2 @@
 Hello git walon
+testing in progress ....
```

Using Command `git diff HEAD`

- `git status -vv` shows both staged and unstaged changes.
- It clearly indicates which changes are staged for commit.
- It also shows which changes are not staged.
- Useful for reviewing file status before committing.

```
MINGW64/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git status -vv
warning: LF will be replaced by CRLF in bw.txt.
The file will have its original line endings in your working directory
warning: LF will be replaced by CRLF in d.txt.
The file will have its original line endings in your working directory
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   bw.txt

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   bw.txt
    modified:   d.txt

Changes to be committed:
diff --git c/bw.txt i/bw.txt
new file mode 100644
index 0000000..85713a6
--- /dev/null
+++ i/bw.txt
@@ -0,0 +1 @@
+hello git users
-----
Changes not staged for commit:
diff --git i/bw.txt w/bw.txt
index 85713a6..36878e1 100644
--- i/bw.txt
+++ w/bw.txt
@@ -1 +1 @@
+hello git users
+hello git users hello
diff --git i/d.txt w/d.txt
index 93541f9..a1c7140 100644
--- i/d.txt
+++ w/d.txt
@@ -1,2 +1,3 @@
 Hello git walon
+testing in progress ....
+grg apka apna yrr ...
```

Using `git status -vv`

Here all the changes which are under - sign are the changes that are not staged for commit.

Showing Differences for a Specific File or Directory :-

git diff file_name

- It shows changes between the previous commit and the locally modified file.
- These changes are not yet staged.
- Lines marked with - indicate unstaged changes.
- Useful for reviewing edits before staging.

```
MINGW64:/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff d.txt
warning: LF will be replaced by CRLF in d.txt.
The file will have its original line endings in your working directory
diff --git a/d.txt b/d.txt
index 93541f9..a1c7140 100644
--- a/d.txt
+++ b/d.txt
@@ -1,2 +1,3 @@
 Hello git walon
-
+testing in progress ....
+GFG apka apna yrr ...
```

Using git diff file_name

- It also works for directories.
- It shows changes between the previous commit and locally modified files in that directory.
- These changes are not yet staged.
- Green lines indicate unstaged changes.

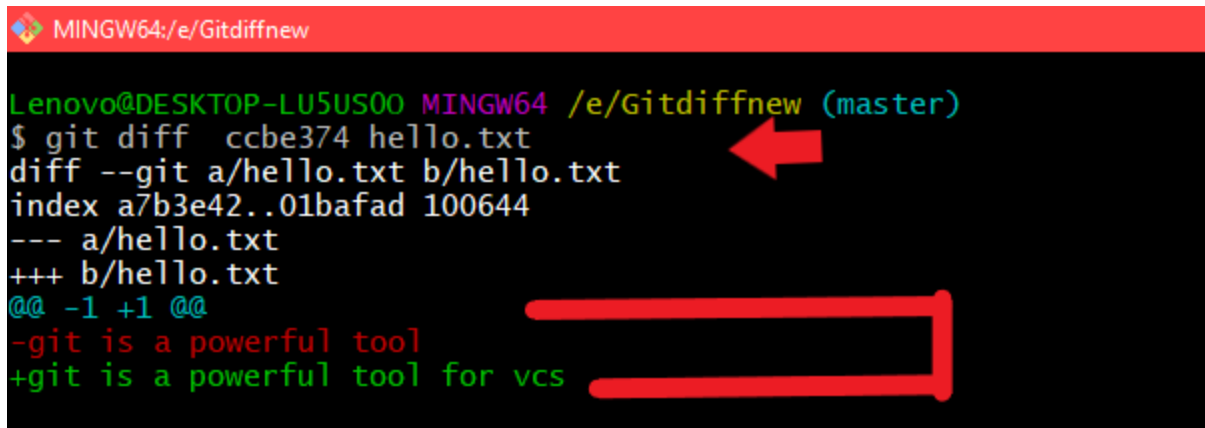
```
MINGW64:/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew/myfolder (master)
$ git diff myfolder
fatal: ambiguous argument 'myfolder': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew/myfolder (master)
$ cd -
/e/Gitdiffnew

Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff myfolder
warning: LF will be replaced by CRLF in myfolder/a.txt.
The file will have its original line endings in your working directory
warning: LF will be replaced by CRLF in myfolder/b.txt.
The file will have its original line endings in your working directory
warning: LF will be replaced by CRLF in myfolder/c.txt.
The file will have its original line endings in your working directory
diff --git a/myfolder/a.txt b/myfolder/a.txt
index e965047..43d44aa 100644
--- a/myfolder/a.txt
+++ b/myfolder/a.txt
@@ -1 +1 @@
-Hello
+Hello Geeks for Geeks
diff --git a/myfolder/b.txt b/myfolder/b.txt
index 457b623..1ad4b0c 100644
--- a/myfolder/b.txt
+++ b/myfolder/b.txt
@@ -1 +1 @@
-Guys
+Guys kya haal hai
diff --git a/myfolder/c.txt b/myfolder/c.txt
index 8630fe9..a084ca8 100644
--- a/myfolder/c.txt
+++ b/myfolder/c.txt
@@ -1 +1 @@
-what is up
+what is up geeks
```

Using git diff folder_name

- It shows the difference between a file in a specific commit and the local HEAD version.
- You specify the commit ID to compare.
- The local HEAD version represents the most recent state of the file.
- This helps track how a file has changed over time.



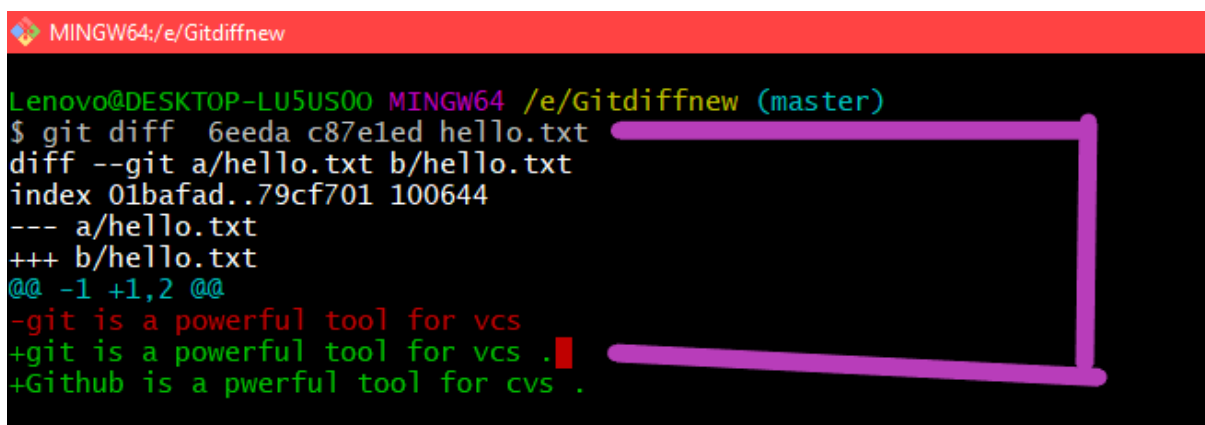
```

MINGW64:/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff ccbe374 hello.txt
diff --git a/hello.txt b/hello.txt
index a7b3e42..01bafad 100644
--- a/hello.txt
+++ b/hello.txt
@@ -1,1 @@
-git is a powerful tool
+git is a powerful tool for vcs

```

Using git diff commit_id file_name

- Green lines show the local HEAD version (current state of the file).
- This represents the most recent changes.
- To compare a file between two commits, use:
git diff commit-id-1 commit-id-2 file_name
- It shows how that file changed between the two commits.



```

MINGW64:/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff 6eeda c87e1ed hello.txt
diff --git a/hello.txt b/hello.txt
index 01bafad..79cf701 100644
--- a/hello.txt
+++ b/hello.txt
@@ -1,2 @@
-git is a powerful tool for vcs
+git is a powerful tool for vcs
+Github is a powerful tool for cvs

```

Using the command git diff commit_id1 commit_id2 file_name

- Green lines show changes between the two specified commits.
- It displays differences for the given commit IDs.
- To compare a commit with the latest branch version for a directory, use:
git diff commit-id branch_name directory_name/
- This shows how the directory changed from that commit to the current branch state.

```
MINGW64:/e/Gitdiffnew

Lenovo@DESKTOP-LU5U500 MINGW64 /e/Gitdiffnew (master)
$ git diff 6d360da master myfolder/
diff --git a/myfolder/a.txt b/myfolder/a.txt
index e965047..43d44aa 100644
--- a/myfolder/a.txt
+++ b/myfolder/a.txt
@@ -1,1 @@
-Hello
+Hello Geeks for Geeks
diff --git a/myfolder/b.txt b/myfolder/b.txt
index 457b623..1ad4b0c 100644
--- a/myfolder/b.txt
+++ b/myfolder/b.txt
@@ -1,1 @@
-Guys
+Guys kya haal hai
diff --git a/myfolder/c.txt b/myfolder/c.txt
index 8630fe9..a084ca8 100644
--- a/myfolder/c.txt
+++ b/myfolder/c.txt
@@ -1,1 @@
-What is up
+What is up geeks
```

Using the command `git diff commit-id branch_name folder_name`

- To show differences between the current and last commit, use:
`git diff HEAD^ HEAD`
- It compares the previous commit with the current commit.
- It shows all changes introduced in the latest commit.

```
MINGW64:/e/Gitdiffnew

Lenovo@DESKTOP-LU5U500 MINGW64 /e/Gitdiffnew (master)
$ git diff HEAD^ HEAD
diff --git a/myfolder/a.txt b/myfolder/a.txt
index e965047..43d44aa 100644
--- a/myfolder/a.txt
+++ b/myfolder/a.txt
@@ -1,1 @@
-Hello
+Hello Geeks for Geeks
diff --git a/myfolder/b.txt b/myfolder/b.txt
index 457b623..1ad4b0c 100644
--- a/myfolder/b.txt
+++ b/myfolder/b.txt
@@ -1,1 @@
-Guys
+Guys kya haal hai
diff --git a/myfolder/c.txt b/myfolder/c.txt
index 8630fe9..a084ca8 100644
--- a/myfolder/c.txt
+++ b/myfolder/c.txt
@@ -1,1 @@
-What is up
+What is up geeks
```

Using `git diff HEAD^ HEAD`

- Patch-compatible diff is used to create a patch file from changes.
- Command: `git diff --no-prefix > some_file.patch`
- The `>` symbol saves the output to `some_file.patch`.
- The patch file contains both staged and unstaged changes.
- Green lines indicate the recently made changes.
- It works similarly to `git diff` but in a patch format for applying elsewhere.

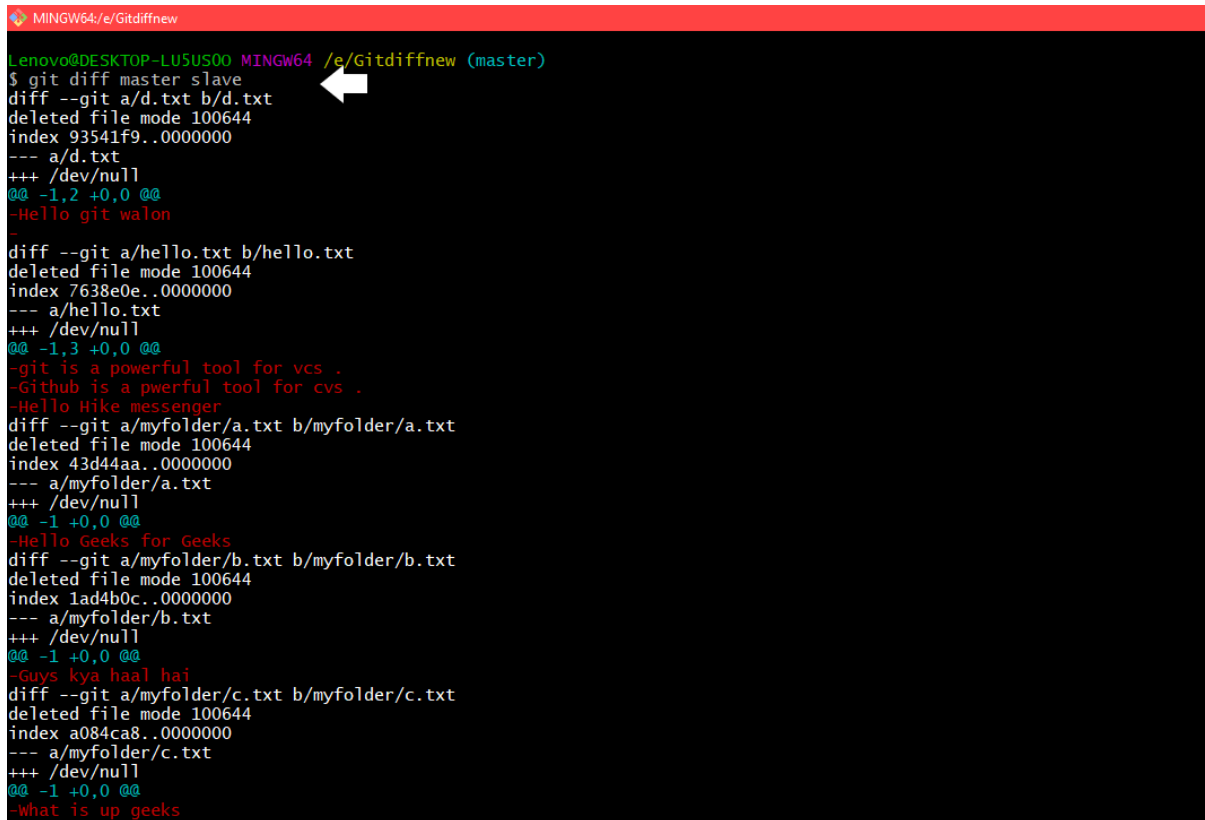
```
MINGW64:/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff --no-prefix < some_file.patch
warning: LF will be replaced by CRLF in d.txt.
The file will have its original line endings in your working directory
diff --git d.txt d.txt
index 93541f9..a1c7140 100644
--- d.txt
+++ d.txt
@@ -1,2 +1,3 @@
Hello git walon
-
+testing in progress ....
+GFG apka apna yrr ...
```

Using git diff --no-prefix > file_name.patch

Difference between two Commit or Branch :-

To view the difference between the two branches we use the following command:

`git diff branch1_name branch2_name`



```
MINGW64:/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff master slave
diff --git a/d.txt b/d.txt
deleted file mode 100644
index 93541f9..0000000
--- a/d.txt
+++ /dev/null
@@ -1,2 +0,0 @@
-Hello git walon
-
diff --git a/hello.txt b/hello.txt
deleted file mode 100644
index 7638e0e..0000000
--- a/hello.txt
+++ /dev/null
@@ -1,3 +0,0 @@
-git is a powerful tool for vcs .
-Github is a powerful tool for cvs .
-Hello Hike messenger
diff --git a/myfolder/a.txt b/myfolder/a.txt
deleted file mode 100644
index 43d44aa..0000000
--- a/myfolder/a.txt
+++ /dev/null
@@ -1 +0,0 @@
-Hello Geeks for Geeks
diff --git a/myfolder/b.txt b/myfolder/b.txt
deleted file mode 100644
index 1ad4b0c..0000000
--- a/myfolder/b.txt
+++ /dev/null
@@ -1 +0,0 @@
-Guys kya haal hai
diff --git a/myfolder/c.txt b/myfolder/c.txt
deleted file mode 100644
index a084ca8..0000000
--- a/myfolder/c.txt
+++ /dev/null
@@ -1 +0,0 @@
-What is up geeks
```

Using `git diff branch1_name branch2_name`

To view the difference between two commit id's following command is used:

`git diff commit-id-1 commit-id-2`

```

MINGW64/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff master slave
diff --git a/d.txt b/d.txt
deleted file mode 100644
index 93541f9..0000000
--- a/d.txt
+++ /dev/null
@@ -1,2 +0,0 @@
-Hello git walon
-
diff --git a/hello.txt b/hello.txt
deleted file mode 100644
index 7638e0e..0000000
--- a/hello.txt
+++ /dev/null
@@ -1,3 +0,0 @@
-git is a powerful tool for vcs .
-Github is a pwerful tool for cvs .
-Hello Hike messenger
diff --git a/myfolder/a.txt b/myfolder/a.txt
deleted file mode 100644
index 43d44aa..0000000
--- a/myfolder/a.txt
+++ /dev/null
@@ -1 +0,0 @@
-Hello Geeks for Geeks
diff --git a/myfolder/b.txt b/myfolder/b.txt
deleted file mode 100644
index 1ad4b0c..0000000
--- a/myfolder/b.txt
+++ /dev/null
@@ -1 +0,0 @@
-Guys kya haal hai
diff --git a/myfolder/c.txt b/myfolder/c.txt
deleted file mode 100644
index a084ca8..0000000
--- a/myfolder/c.txt
+++ /dev/null
@@ -1 +0,0 @@
-What is up geeks

```

Using command git diff commit-id-1 commit-id-2

To view difference with current branch

git diff name_of_branch

```

Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (slave)
$ git diff master
diff --git a/d.txt b/d.txt
deleted file mode 100644
index 93541f9..0000000
--- a/d.txt
+++ /dev/null
@@ -1,2 +0,0 @@
-Hello git walon
-
diff --git a/hello.txt b/hello.txt
deleted file mode 100644
index 7638e0e..0000000
--- a/hello.txt
+++ /dev/null
@@ -1,3 +0,0 @@
-git is a powerful tool for vcs .
-Github is a pwerful tool for cvs .
-Hello Hike messenger
diff --git a/myfolder/a.txt b/myfolder/a.txt
deleted file mode 100644
index 43d44aa..0000000
--- a/myfolder/a.txt
+++ /dev/null
@@ -1 +0,0 @@
-Hello Geeks for Geeks
diff --git a/myfolder/b.txt b/myfolder/b.txt
deleted file mode 100644
index 1ad4b0c..0000000
--- a/myfolder/b.txt
+++ /dev/null
@@ -1 +0,0 @@
-Guys kya haal hai
diff --git a/myfolder/c.txt b/myfolder/c.txt
deleted file mode 100644
index a084ca8..0000000
--- a/myfolder/c.txt
+++ /dev/null
@@ -1 +0,0 @@

```

Using git diff branch_name

To view the difference with commit_id

git diff commit id

```
MINGW64:/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (slave)
$ git diff a6d6c45
diff --git a/d.txt b/d.txt
deleted file mode 100644
index 93541f9..0000000
--- a/d.txt
+++ /dev/null
@@ -1,2 +0,0 @@
-Hello git walon

diff --git a/hello.txt b/hello.txt
deleted file mode 100644
index 7638e0e..0000000
--- a/hello.txt
+++ /dev/null
@@ -1,3 +0,0 @@
-git is a powerful tool for vcs .
-Github is a powerful tool for cvs .
-Hello Hike messenger

diff --git a/myfolder/a.txt b/myfolder/a.txt
deleted file mode 100644
index 43d44aa..0000000
--- a/myfolder/a.txt
+++ /dev/null
@@ -1 +0,0 @@
-Hello Geeks for Geeks

diff --git a/myfolder/b.txt b/myfolder/b.txt
deleted file mode 100644
index 1ad4b0c..0000000
--- a/myfolder/b.txt
+++ /dev/null
@@ -1 +0,0 @@
-Guys kya haal hai

diff --git a/myfolder/c.txt b/myfolder/c.txt
deleted file mode 100644
index a084ca8..0000000
--- a/myfolder/c.txt
+++ /dev/null
@@ -1 +0,0 @@
-What is up geeks
```

Using git diff commit_id

To view the summary of changes

git diff --stat branch or we can write git diff -stat commit_id

```
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff --name-only bc8a7f9 myfolder/
myfolder/goku.txt
myfolder/naruto.txt

MINGW64:/e/Gitdiffnew
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (slave)
$ git diff --stat master
 d.txt          | 2  --
 hello.txt      | 3  ---
 myfolder/a.txt | 1  -
 myfolder/b.txt | 1  -
 myfolder/c.txt | 1  -
 new.txt        | 1  +
 some_file.patch | 23 -----
 7 files changed, 1 insertion(+), 31 deletions(-)
```

Using command git diff -stat branch to view the summary of changes in the branch

To view files that changed after a certain commit

git diff --name-only commit-id

```
MINGW64:/e/Gitdiffnew

Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ ls
at.txt  bw.txt  d.txt  d.txt.orig  d.txt.rej  hello.txt  myfolder/  some_file.patch  some_file.patch.orig

Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ cat bw.txt
hello git users Contact to google dev

Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff 5d79a58
warning: LF will be replaced by CRLF in bw.txt.
The file will have its original line endings in your working directory
diff --git a/bw.txt b/bw.txt
index 85713a6..925f655 100644
--- a/bw.txt
+++ b/bw.txt
@@ -1,1 @@
-hello git users
+hello git users Contact to google dev
```

Using git diff --name-only commit-id command

To view files that are different than a branch

git diff --name-only branch_name

```
MINGW64:/e/Gitdiffnew

Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff --name-only master
github.txt
newfile.txt
```

Using git diff --name-only branch_name

To view files in a folder that changed after a commit

git diff --name-only commit-id folder-path

```
Lenovo@DESKTOP-LU5US00 MINGW64 /e/Gitdiffnew (master)
$ git diff --name-only bc8a7f9 myfolder/
myfolder/goku.txt
myfolder/naruto.txt
```

Using the command git diff --name-only commit-id folder-path

Differences Between HEAD, Working Tree, and Index :-

Component	Description	Key Commands
HEAD	Pointer to the latest commit in the current branch	git log -1
Working Tree	The directory where your project files are checked out	git status
Index	Intermediate area to prepare changes for the next commit	git add <filename>, git diff --cached